UNIT 4 · 8 hours

Rules for Construction of the Root Locus



Stability of Systems

Control Systems (22EEG53)
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Learning objectives

- Apply the rules for plotting a root locus
- Find asymptotes, centroid, breakaway points and angle of departure
- Find the jω-axis crossing using Routh
- Sketch the root locus for a third-order system

Syllabus context

Stability of systems: characteristic equation, location of roots, absolute and relative stability, Routh–Hurwitz criterion (necessary and sufficient conditions), root locus concept, rules for construction of root locus.

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Rules (1–5)

Rule	Statement
1 Symmetry	Locus is symmetric about the real axis.
2 Start/end	Branches start at open-loop poles (K = 0) and end at zeros (K = ∞), or at infinity.
3 Number of branches	Equals max(P, Z), P = poles, Z = zeros.
4 Real axis	A point is on the locus if the number of poles + zeros to its right is odd.
5 Asymptotes	N = P − Z asymptotes; angles $(2q + 1)180°/(P − Z)$; centroid $\sigma A = (\Sigma poles − \Sigma zeros)/(P − Z)$.

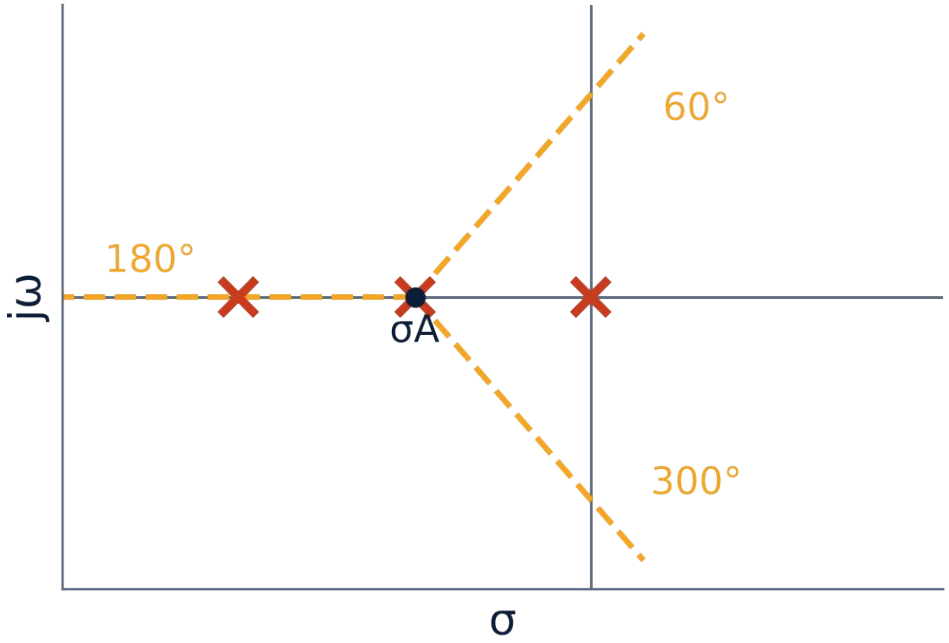
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Rules (6–8)

Rule	Statement
6 Breakaway / break-in	Solve dK/ds = 0 with K = −1/G1H1(s); valid roots lie on the locus.
7 jω-axis crossing	Use Routh's array (K and ω at the marginal row) or put s = jω in 1 + GH = 0.
8 Angle of departure / arrival	φd = 180° − Σ(angles from other poles) + Σ(angles from zeros) at a complex pole.
9 Gain at a point	K = Π|distances to poles| / Π|distances to zeros|.

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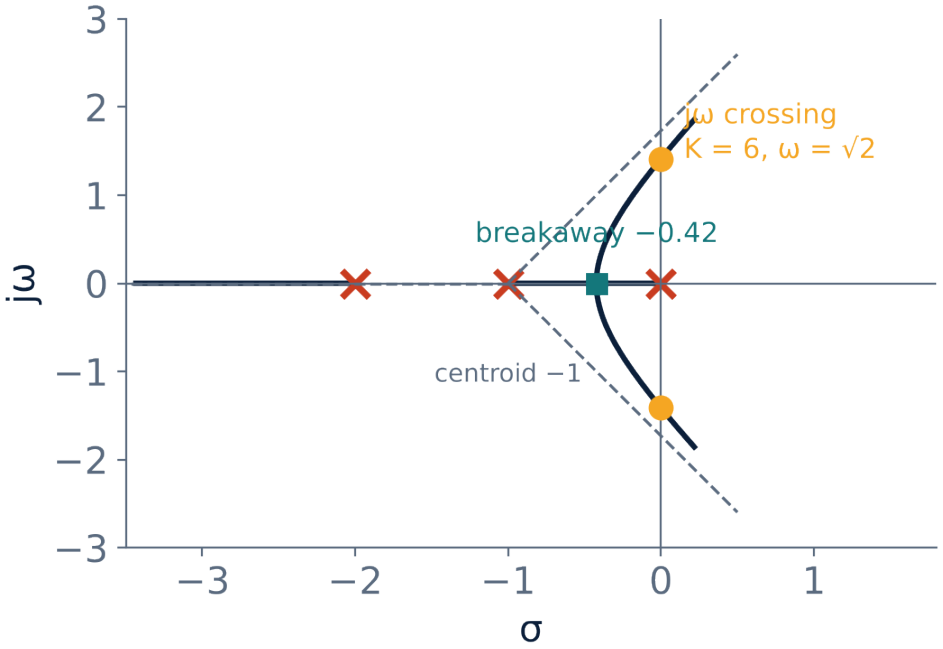
Asymptotes: K/[s(s+1)(s+2)]



- P = 3, Z = 0: three asymptotes.
- Angles: 60°, 180°, 300°.
- Centroid σA = (0 − 1 − 2)/3 = −1.
- Real-axis locus: 0 to −1 and −2 to −∞.

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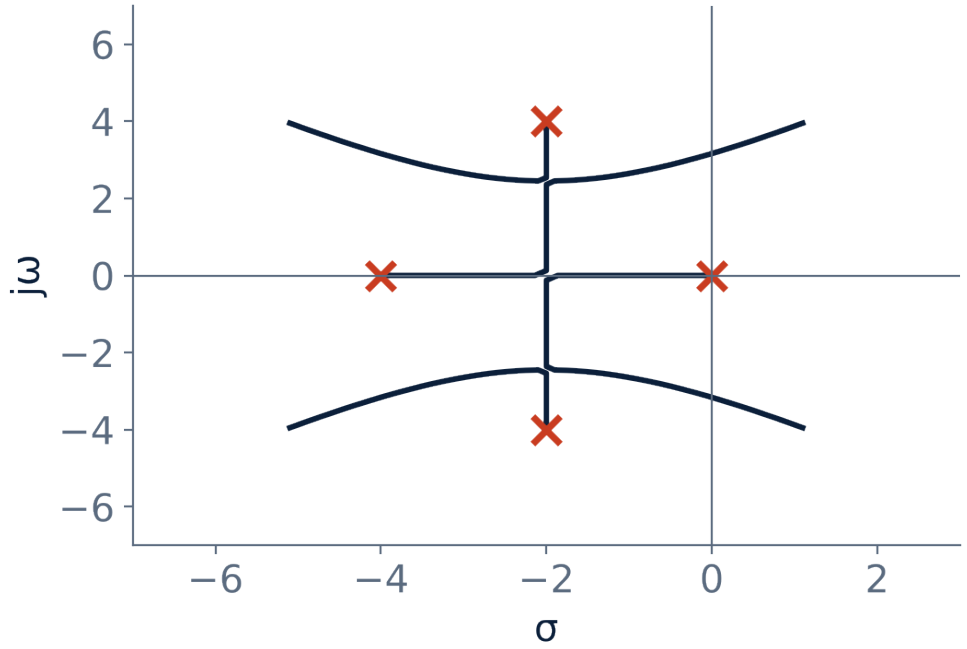
Completed root locus



- Breakaway: $K = -s(s+1)(s+2)$; $dK/ds = 0 \Rightarrow 3s^2 + 6s + 2 = 0 \Rightarrow s = -0.423$ (on locus), -1.577 (not).
- jω crossing: Routh gives $K = 6$, $\omega = \sqrt{2}$.
- $K > 6$: two roots in the RHP: unstable.
- Gain at breakaway: $K = 0.385$.

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Fourth-order example: $K/[s(s+4)(s^2+4s+20)]$



- Poles: 0, −4, −2 ± j4.
- Asymptotes: 45°, 135°, 225°, 315°; σA = −2.
- Breakaway: −2 and −2 ± j2.45.
- Angle of departure at −2 + j4: 180° − (116.6° + 63.4° + 90°) = −90°.
- jω crossing: K = 260, ω = 3.16.

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Numerical: the jω crossing

Problem Characteristic equation $s^3 + 3s^2 + 2s + K = 0$. Find K and ω at the imaginary-axis crossing.

- Routh: s^1 element $(6 - K)/3 = 0 \Rightarrow K = 6$
- Auxiliary equation from the s^2 row: $3s^2 + K = 0 \Rightarrow 3s^2 + 6 = 0$
- $s = \pm j\sqrt{2}$

Answer: K = 6, ω = 1.414 rad/s

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Key takeaways

- Branches start at poles, end at zeros/infinity; symmetric about the real axis
- Asymptotes: angles $(2q+1)180°/(P-Z)$, centroid = $(\Sigma p - \Sigma z)/(P-Z)$
- Breakaway: $dK/ds = 0$; $j\omega$ crossing: Routh
- Angle of departure from a complex pole: $180° - \Sigma\varphi p + \Sigma\varphi z$

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Practice questions

Q1. Sketch the root locus of K/[s(s+3)(s+6)] and find the marginal K.

Q2. Find asymptote angles and centroid for K(s+1)/[s(s+2)(s+4)].

Q3. Find the breakaway point of K/[s(s+4)].

Q4. State the rule for real-axis segments.